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FIFTH REVIEW UNDER THE EXTENDED CREDIT FACILITY ARRANGEMENT—DEBT SUSTAINABILITY ANALYSIS

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Madagascar ***Joint Bank-Fund Debt Sustainability Analysis***	
Risk of external debt distress	***Low***
Overall risk of debt distress	***Moderate***
Granularity in the risk rating	***n/a***
Application of judgment	***No***

Madagascar is assessed at low risk of external debt distress. This marks a change from moderate risk in the June 2018 DSA, despite a broader definition of external debt, and reflects an upgrade in Madagascar's debt carrying capacity rather than a change in the debt path. Under the baseline, external public and publicly guaranteed (PPG) debt is well below applicable thresholds. Stress tests do not breach the threshold applicable to countries with medium debt-carrying capacity. Total (external plus domestic) PPG debt is below the benchmark under the baseline, but growth shocks drive the present value of the ratio of debt to GDP above the benchmark. Shocks could also introduce liquidity problems, as the debt-service to revenue ratio could exceed 100 percent over the long term. The overall rating, of moderate debt distress, remains consistent with the 2018 DSA. These assessments continue to be supportive of Madagascar's current plans to scale up its borrowing to meet its investment needs, though other factors are also critical.

[1] Prepared by the IMF and the World Bank. This DSA follows the Guidance Note of the Join Bank-Fund Debt Sustainability Framework for Low Income Countries, February 2018. This new framework builds on the 2013 framework used for the previous Madagascar DSA, which was published in June 2018 (IMF Country Report No. 18/239).

PUBLIC DEBT COVERAGE

1. This DSA expands debt coverage of public and publicly guaranteed external and domestic debt to include State-owned Enterprises' domestic debt as well as central bank external liabilities. This marks expanded coverage since the previous DSA (Text Table 1). PPG debt now includes all external liabilities held by the central bank (4 per-cent of GDP in 2018) including all borrowing from the IMF. Following recently-secured historical estimates, we now include debts owed by state-owned enterprises (SOEs) in cases where the government has at least 50 percent of the shares. These debts, which amount to an estimated 6 percent of GDP and are mostly those of the utility Jirama, are in addition to the coverage of direct guarantees provided by the central government. The DSA does not include the debt of local government owing primarily to data limitations, although anecdotal discussion suggests borrowing typically does not occur at this level and should require centralized approval. The measure of debt is on a *gross* rather than *net* basis and the currency criterion is used to distinguish between domestic and external debt.[2]

Text Table 1. Madagascar: Public Debt Coverage Under the Baseline Scenario

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	
3	Other elements in the general government	
4	o/w: Social security fund	
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

Text Table 2. Madagascar: Coverage of the Contingent Liabilities' Stress Test

The country's coverage of public debt	The central government, central bank, government-guaranteed debt, non-guaranteed SOE debt	
	Default	Used for the analysis
Other elements of the general government not captured in 1.	0 percent of GDP	0.0
SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2.0
PPP	35 percent of PPP stock	0.0
Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	5.0
Total (2+3+4+5) (in percent of GDP)		7.0

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

[2] Locally-issued debt denominated in local currency held by non-residents and/or locally-issued debt denominated in foreign currency held by residents is likely insignificant. Owing to limitations in available data, the results would be the same if selecting the residency criterion.

2. The coverage assumed for contingent liabilities is a shock of 7 percent of GDP. This reflects the default setting for SOEs, while minimum allowed values are used for PPPs and financial markets (Text Table 2).

- Estimated debt for SOEs in which the government has a majority stake are included in the baseline as domestic debt. However, government recognition of some of those liabilities could require external financing. Other potential contingencies include future recapitalization of the postal savings scheme and the Madagascar Savings Fund (*Caisse d'Epargne de Madagascar, CEM*), which would likely amount to less than 1 per-cent of GDP. Following the recapitalization of Air Madagascar and its strategic partnership with Air Austral, no additional public support to Air Madagascar beyond the guarantee included in the baseline debt is expected, but an implicit government guarantee of future airline obligations is a risk.

- Exposures through PPPs are set to zero since, as per the World Bank's PPP database, PPPs comprise less than ½ percent of GDP. The authorities plan to develop more PPPs in future.

- The default minimum value of 5 percent is programmed for financial markets. Most banks are financially solid with deposits exceeding loans and majority foreign shareholders. Dollarization of deposits and credits is not pronounced, and banks' foreign assets generally exceed their foreign liabilities.

BACKGROUND

A. Recent Debt Revisions and Developments

3. PPG debt ratios are lower than at the start of the ECF program. Using the expanded definitions, total public debt rose to 48.4 percent of GDP in 2015 and declined to 45.7 percent in 2018, reflecting stronger revenue collection, rising nominal GDP growth, and an appreciating real effective exchange rate. PPG debt was broadly unchanged since 2017 (46 percent), with external PPG debt rising by 1 percent of GDP and domestic debt declining by 1.3 percent. The 2018 DSA projected debt of 35 percent of GDP for 2018; the higher value in this DSA is mostly accounted for by the now-expanded coverage (Table 1, Figure 1).

4. Private external and hence total debt estimates have been revised upwards, and debt service has been revised downwards. Staff sourced updated estimates for external private debt drawing on confidential mining sector information in its September 2018 mission. Previous external debt estimates were based in part on fast amortization rates. Slower historical and forecast debt service reflect information attained during the mission.[3] This has resulted in upward revisions to historical private debt levels. For example, estimates for 2018 stand at 27.5 percent of GDP, which is substantially higher than the 11.5 per-cent estimated in the 2018 DSA. Nonetheless, private debt data remains incomplete. The authorities are working to reduce debt data gaps as they improve various components of external sector statistics by drawing on IMF technical assistance. Total public and private external debt is estimated at 58 percent of GDP in 2018, compared to 37 percent in the 2018 DSA but down 3 percent of GDP since 2016.

[3] We also expanded coverage to include commercial banks (worth 0.6 percent of GDP in 2018).

Text Table 3. Madagascar: Breakdown of Total PPG Debt (2015-18)

Creditor	2015	2016	2017	2018
		Amount (US$m)		
Domestic debt, *of which:*	**1,689**	**1,682**	**1,827**	**1,763**
Securities inc. BTA, BTF, BTS[1]	370	526	719	726
Debt to the Central Bank	386	337	297	272
Arrears	346	210	146	71
Other inc. SOE debt	587	610	665	693
External debt, *of which:*	**2,816**	**2,845**	**3,262**	**3,549**
Multilateral	2,006	2,052	2,276	2,368
Paris Club	146	137	165	189
Non-Paris Club	356	324	290	308
Commercial & Gauranteed	25	23	83	204
External debt of the Central Bank	282	310	449	480
Total PPG debt	**4,505**	**4,528**	**5,089**	**5,312**
		Percent of GDP		
Domestic debt, *of which:*	**16.8**	**17.5**	**16.5**	**15.2**
Securities inc. BTA, BTF, BTS	4.2	5.5	6.5	6.3
Debt to the Central Bank	4.3	3.5	2.7	2.3
Arrears	3.9	2.2	1.3	0.6
Other inc. SOE debt	6.6	6.3	6.0	6.0
External debt, *of which:*	**31.7**	**29.6**	**29.5**	**30.5**
Multilateral	22.6	21.4	20.6	20.4
Paris Club	1.6	1.4	1.5	1.6
Non-Paris Club	4.0	3.4	2.6	2.6
Commercial & Gauranteed	0.3	0.2	0.7	1.8
External debt of the Central Bank	3.2	3.2	4.1	4.1
Total PPG debt	**48.4**	**47.1**	**46.0**	**45.7**
		Percent of total		
Domestic debt, *of which:*	**37.5**	**37.2**	**35.9**	**33.2**
Securities inc. BTA, BTF, BTS	8.2	11.6	14.1	13.7
Debt to the Central Bank	8.6	7.4	5.8	5.1
Arrears	7.7	4.6	2.9	1.3
Other inc. SOE debt	13.0	13.5	13.1	13.0
External debt, *of which:*	**62.5**	**62.8**	**64.1**	**66.8**
Multilateral	44.5	45.3	44.7	44.6
Paris Club	3.3	3.0	3.2	3.6
Non-Paris Club	7.9	7.1	5.7	5.8
Commercial & Gauranteed	0.5	0.5	1.6	3.8
External debt of the Central Bank	6.3	6.8	8.8	9.0
Total PPG debt	**100.0**	**100.0**	**100.0**	**100.0**

Sources: Malagasy authorities; and staff estimates.

[1]BTA are Treasury bills with less than one year maturity; BTF and BTS are Treasury bonds with maturity ranging from 1 to 3 years.

5. External sources account for two-thirds of PPG debt (text table 3). The contribution of external debt has been rising following Madagascar's re-engagement with the international community. Contributors to the rise are chiefly multilateral sources, IMF lending to the Central Bank, and commercial loans. As at end-2018, two-thirds of external debt is owed to multilateral creditors, in particular the World Bank and African Development Bank, on highly concessional terms including long maturities. Non-Paris club creditors account for 9 percent of external debt, which is a lower share than at the start of the program. Domestic debt declined in 2018, mostly due to continued progress in reducing arrears. Government securities are the largest category, followed by estimated debts of SOEs in which the government has a majority stake.

B. Macroeconomic Assumptions

6. DSA projections remain consistent with the authorities' plan to scale up much needed infrastructure investment, albeit at a slower pace than in the previous DSA.

- As per the authorities' plans, external project loans signed, being studied, or being negotiated amount to approximately $1.6 billion in nominal terms for 2018-19, compared with $1.4 billion planned in the 2018 DSA, reflecting more specific project proposals. This includes $0.5 billion already contracted in 2018, which is $0.4 billion less than the authorities' plans used for the 2018 DSA. The authorities' plans therefore entail an ambitious acceleration in contracting for 2019. In particular, projected external borrowing for 2019 is $1 billion in nominal terms and $0.6 billion in PV terms. In the first five months of the year, only one new loan worth $ 150 million was signed.

- Actual and planned disbursements lag loans contracted. Disbursements, worth $0.25 billion in nominal terms in 2018 (2 percent of GDP), were $0.1 billion lower than forecast in the 2018 DSA owing to weak execution. Staff's projections assume under execution relative to the authorities' plans. We assume an increase to almost $0.4 billion in 2019. Disbursements beyond 2019 introduce some catch up to account for lower-than-anticipated historical disbursements, but at a slower pace than the authorities' plans, with a lower peak, and with the scaled-up phase extending further into the medium term than in the 2018 DSA. The 2019-21 debt strategy document indicates financing assumptions broadly consistent with the 2018 DSA. Over the long term, we assume disbursements fall gradually as a share of GDP back to around the share forecast for 2019 (around 3 percent of GDP). Thus, the assumption is that Madagascar will continue to borrow over the long run for development but at a slower pace than the scaled-up pace envisaged for the medium term.

7. Changes in short- and medium-term macroeconomic assumptions since the DSA 2018 largely reflect observed outcomes for 2018 (text table 4)**.** GDP growth projections remain broadly unchanged. Lower inflation forecasts are in part attributable to lower than expected inflation in 2018. The non-interest current account deficit has been revised down over the short term as vanilla and mining proceeds have again exceeded expectations. Over the next 5 years, vanilla prices are now expected to fall by one third, contributing to a steeper decline in the current account balance compared to the 2018 DSA.[4] However, this is offset by upward revisions to short- and medium-run mining export volumes following private sector discussions, and the 2029 deficit remains similar to the 2018 DSA. The 2019 primary deficit[5] is lower than in the 2018 DSA, reflecting lower expenditure estimates following budget discussions with the authorities. Revenues are revised down over the medium term to reflect more conservative projections for tax revenue showing a steady but credible upward trend. Some budget grant disbursements have been postponed to 2019 and 2020. Since we continue to include only firm commitments to be conservative, the assumptions include a steep decline in budget grants in 2020 and again (to zero) thereafter. Thus, assumed (project and budget) grants decline substantially. Expenditure has been revised down over the short run mainly due to observed under execution of investment spending.

[4] This time frame is consistent with estimates of supply responses from elsewhere in the world, and the steeper reduction also prompts a downward revision to assumed vanilla production growth over the longer term.

[5] The primary deficit in the table includes foreign financed capital expenditure and grants, which are excluded from IMF program quantitative targets.

Text Table 4. Madagascar: Baseline Macroeconomic Assumptions for DSA

(In percent of GDP, unless otherwise indicated)	2019		2020		2024		2029	
	2018 DSA	new DSA	old DSA	new DSA	old DSA	new DSA	old DSA	new DSA
Real GDP growth (percent)	5.4	5.2	5.3	5.3	4.8	4.8	4.7	4.6
Inflation, average (percent)	7.2	6.7	6.3	6.3	5.5	5.0	5.2	5.0
Non-interest CA deficit	2.4	1.3	3.4	2.8	2.0	3.8	3.1	2.9
Primary deficit	3.6	1.3	4.4	3.5	2.0	4.5	1.2	2.4
Total revenues, excluding grants	12.5	12.5	13.0	12.6	14.3	13.7	15.3	14.5
Grants	2.7	3.9	1.4	2.4	1.4	0.9	1.2	0.7
Non-Interest Expenditure	18.8	17.7	18.8	18.5	17.7	19.1	17.7	17.6

Sources: Malagasy authorities, World Bank and IMF.

8. Long-term assumptions introduce additional conservatism and incorporate base effects from 2018 outturns. Since average GDP growth rates typically taper as countries grow, long-term GDP growth is projected to slow slightly towards 4¼ percent over the projection period. Risks from recurrent natural disasters are implicitly incorporated in the baseline. The non-interest current account deficit is projected at around 3 percent in 2029 and 2½ percent in 2039. This marks a flatter profile in the outer years than in the previous DSA, which had a deficit rising to 4 percent in 2038. This DSA has more conservative assumptions on exports, including mining (following updated information on project lifespans) and processing zone imports/exports (to assume a more moderate share of total trade). By the end of the projection period, this change is offset by lower imports compared to the 2018 DSA. Other current account flows are higher, in part due to higher base effects from 2017-8.[6] Project grants are revised downward slightly since some sources of support are contingent on the external debt distress rating. We have assumed a slightly lower nominal exchange rate depreciation in order to be consistent with a very small real appreciation each year, which is in line with guidance and other country experience. The primary fiscal deficit is estimated at around 2½ percent in 2029, which is higher than in the 2018 DSA. This primarily mostly reflects more cautious assumptions on revenue growth.

9. This outlook remains subject to risks, many tilted to the downside. Lower than projected public investment would reduce borrowing but weigh negatively on growth. Losses at SOEs could crowd out pro-growth priority spending, while insufficient progress in fighting corruption could harm confidence. Madagascar also remains highly vulnerable to long-run terms-of-trade variations from the baseline as well as short-term fluctuations. Key drivers of the terms of trade include prices for imports of oil as well as exports of vanilla, which could return to their long-run average faster than forecasts, and metals, which can be particularly susceptible to growth in key global economies. More generally, negative external developments, such as lower global growth, rising protectionism and a retreat from multilateralism, could reduce exports and donor assistance. At the same time, there are upside risks for growth, since projections are conservative on FDI and the multiplier effect from scaled-up public investment, and to the terms of trade. Revenue projections could prove unduly conservative.

[6] This revision also generates a smoother path for the residual between debt accumulation and identified debt creating flows, which fell abruptly to zero in the previous DSA.

C. Drivers of Debt Dynamics

10. Over the medium term, scaling up of foreign-financed public investment drives an increase in debt compared with the 5-year average (Figure 3). In 2014-8, external debt rose substantially slower than anticipated in the 2015 DSA. The difference is clearer when comparing the 2015 and 2018 vintages, since this DSA introduces an expanded definition of PPG debt. A key contributor is an unanticipated more than ten-fold rise in Vanilla prices, which boosted the current account and the Ariary. Following re-engagement with the international donor community and the IMF program in 2016, external debt is expected to rise, permitting imports to support much-needed investment. The expected pace of project execution has been revised down slightly relative to the 2018 DSA, leading to a marginally more gradual rise in external debt. Other contributors, such as FDI and a residual that includes unrepatriated mining receipts and reserve accumulation, remain broadly similar to recent history (and unchanged from the 2018 DSA). Public (domestic and external) debt changes were slightly less than those expected in the past, largely reflecting external debt developments. Debt is anticipated to rise slightly (and more slowly than in the past) as the higher primary deficit is almost offset by GDP growth.

11. Government capital spending accounts for much of the expected increase in growth compared to the past (Figure 4). Over the next three years, the primary deficit (including spending financed by external sources) is expected to rise, adding a positive impulse that sustains GDP growth despite the unwinding of the impulse from the 2018 agriculture recovery. The low implied fiscal multiplier may suggest upside risk to growth, as indicated earlier. The contribution of public investment to real GDP growth is expected to be higher than in recent history but lower than in the previous DSA. This reflects the slower-than-anticipated scaling up of private investment, which nonetheless could again be slower than forecast.

D. Country Classification and Determination of Stress Test Scenarios

12. Madagascar's composite indicator has risen and is above the threshold to classify its debt carrying capacity as medium. The upgrade is consistent with incremental progress recorded in the December 2017 World Bank Debt Performance Assessment and summarized in the 2018 DSA. For that DSA, Madagascar had a CPIA[7] score of 3.17, which reflected recent steady improvements but was below the threshold of 3.25 needed for graduation from weak to medium debt-carrying capacity. The CPIA has since risen again. Moreover, the new DSA framework complements the CPIA with import cover of reserves, remittances, Madagascar's real GDP growth, and world GDP growth to form a Composite Indicator (CI) score. The calculation of the CI is based on 10-year averages across 5 years of historical data and 5 years of projection. The April 2019 WEO constitutes a repeated signal to upgrade of the country from weak to medium debt-carrying capacity (Text Table 5). Contributors to debt-carrying capacity should be carefully monitored. Reserves account for the bulk of the increase since the October 2018 WEO. Substantial downward revisions to medium-term reserves projections or global growth alongside reversal of debt-carrying capacity progress could bring Madagascar's CI score closer to the lower threshold.

[7] The Country Policy and Institutional Assessment (CPIA) is an index compiled annually by the World Bank for all IDA-eligible countries. It consists of 16 indicators in four categories: (1) economic management; (2) structural policies; (3) policies for social inclusion and equity; and (4) public sector management and institutions.

Text Table 5. Madagascar: Calculation of Debt-Carrying Capacity

	DSA June 2018	WEO October 2018	WEO April 2019
CPIA 1/	1.22	1.24	1.24
Real growth rate (in percent)		0.12	0.12
Import coverage of reserves (in percent)		0.86	0.91
Remittances (in percent)		0.05	0.05
World economic growth (in percent)		0.48	0.48
Composite Indicator Score		2.75	2.80
Debt Carrying Capacity 2/	**Weak**	**Medium**	**Medium**

1/ CPIA score for June 2018 DSA of 3.17 based on 2014-16 average and rescaled for comparison.

2/ Unscaled CPIA Threshold to reach medium capacity was 3.25 and Composite Indicator Score Threshold for medium capacity is 2.69.

13. The improvement in debt-carrying capacity classification implies an increase in thresholds for debt burdens. For the present value (PV) of external debt relative to GDP, thresholds rise from 30 to 40 percent. Relative to exports, the threshold rises from 140 to 180 percent. The threshold for the PV of external debt service increases from 10 to 15 percent for exports, and from 14 to 18 percent for revenue. The benchmark for the PV of total public debt also rises from 35 to 55 percent of GDP.

14. Stress tests generally follow standardized settings and include shocks for natural disasters and commodity export prices. The contingent liability stress test is based on the quantification of potential contingent liabilities discussed above, and the standardized stress tests apply the default settings. Madagascar remains exposed and vulnerable to natural disaster shocks, like cyclones, and hence qualifies for the natural disaster shock.[8] Since commodities comprise about half of goods and services exports, we also include a commodity shock stress test. The standardized settings of this stress test are customized to better reflect Madagascar's country specific circumstances. In particular, we assume an illustrative fall in prices equivalent to 10 percent of commodity exports, with no mitigating effect on imports, alongside declines in real GDP growth of 0.5 percent and in fiscal revenue of 0.25 percent of GDP. The shock occurs in 2020 and unwinds gradually by 2029. Residual financing for external debt stress tests is assumed to be from external sources at the baseline concessional rates, which is the default setting. For overall public debt stress tests, limited recourse to domestic sources in the short run prompts us to assume 80 percent of additional financing would come from external sources and that the remaining domestic financing would be more expensive than under the baseline.

[8] We apply the default settings for this one-off shock in the template, namely a 10 percentage-point rise in the debt-to-GDP ratio alongside a fall in real GDP growth (1.5 percent) and exports (3.5 percent), in 2020.

DEBT SUSTAINABILITY RESULTS

A. External Debt Sustainability

15. Under the baseline, rising external PPG debt remains comfortably below thresholds (Table 1). It is projected to rise from 30½ percent of GDP in 2018 to 43 percent of GDP in 2029 and decline to about 38 percent of GDP in 2039. The profile using the expanded definition is similar to that in the 2018 DSA, which had external debt rising to 36 percent in 2028 and falling to 32 percent in 2038. Debt-creating flows include a steadily rising current account deficit over the medium term (owing to declines in the trade balance and falling inflows from official transfers) and slightly weaker endogenous debt dynamics (higher interest rates and slowing GDP growth).[9] In present value terms, external debt is projected to rise from 18 percent of GDP in 2018 to almost 30 percent of GDP in 2029 and 27 percent by 2039. The higher relative long-term rise in PV terms is the result of our assumption that borrowing will become less concessional over the long term, reflecting assumed rises in interest rates and a less favorable mix of concessional lenders. Together with expiring grace periods for some loans, this explains why debt service indicators rise substantially off their low base. For example, the projected debt service to exports ratio rises from 2½ in 2018 to 6 in 2029. Nonetheless, all indicators remain well below the applicable thresholds for Madagascar.

16. None of the shock scenarios breach the external thresholds applied for Madagascar by 2029 but would breach or come close to the thresholds for those with weak capacity (Table 3; Figure 1). The most serious shock is that to export growth, which is set to one standard deviation of the projection or historical average (whichever yields lower exports) in 2020 and 2021. Using the applicable thresholds for Madagascar for medium debt carrying capacity, the most serious shock does not breach any of the four thresholds, although the debt-service-to-revenue ratio peaks at about 17, which is close to the threshold of 18 and above the threshold of 14 for weak capacity countries. The PV of debt to GDP rises to 38 percent of GDP in 2025 and remains near 37 percent in 2029, below the indicated threshold of 40 applicable to Madagascar (but above the threshold of 30 that would apply for those with weak capacity). The same shock would drive the PV of debt-to-exports ratio to about 160, which is below the threshold of 180 for Madagascar but above the weak-capacity threshold, and push the debt-service to exports ratio to 11, which is below Madagascar's threshold of 15 but above the weak-capacity threshold of 10.

17. The historical scenarios and granularity of assessment are less applicable to Madagascar. The historical scenario shows the PV of debt to GDP exceeding 40 in 2029 but, as discussed in the 2017 DSA, is currently not a useful stress test or measure of realism since it includes data with atypically high current account deficits related to private mining investment.[10] For countries that breach external debt thresholds, further granularity can be provided by assessing the gap between baseline debt projections and the threshold between moderate and high debt distress. This is not the case for Madagascar given its low risk of debt distress. However, illustrative calculations applying lower thresholds that would be associated with

[9] The residual, which includes reserve accumulation, unrepatriated mining receipts, and potentially other misclassified BOP entries, is assumed to decline steadily over time as the share of mining exports declines and reserve accumulation slows.

[10] For example, excluding 2009 and 2010 from the calculation halves the historical non-interest current account deficit (over the last eight years) to 2½ percent.

countries of a lower debt carrying capacity indicate that, for the PV of debt to GDP and of debt service to revenue, Madagascar's baseline debt burden places it in the zone of having limited space to absorb shocks.[11]

18. Private sector debt is not assessed to pose a significant threat to external sustainability. Private external debt is projected to decline as the loans related to a major mining project are repaid. Using improved information on financial plans, including on mining, the share of private external debt in GDP is anticipated to fall by half within the next decade. Given the exceptional nature of the mining projects, the DSA does not forecast substantial new external borrowing from the private sector over the corresponding period. However, since the 2018 DSA, we have conservatively assumed more borrowing would be needed to sustain mining exports towards the end of the DSA horizon, contributing to private debt equivalent to about 10 percent of GDP in 2039. Such debt is not assessed to pose a significant threat to external sustainability. Consistent with recent experience leading to slower amortization, the ultimate liability for these loans is held by the multinational shareholders rather than resident entities (such as domestic banks or the government).

B. Total Public Debt Sustainability

19. Under the baseline, moderate rises in total domestic and external public debt remain comfortable (Table 2). Total debt is projected to rise from 45½ percent of GDP in 2018 to 57 percent of GDP in 2029 and remain steady thereafter. In the 2018 DSA, which excluded external debts of the central bank and domestic SOE debt, total debt was projected at 44 percent of GDP in 2028 and 2038. The domestic component is projected to decline from 15 percent of GDP in 2018 to 14 percent in 2029 and rise above 18 percent by 2039, reflecting increasing reliance on domestic financing as the sector develops and external support becomes less essential. In PV terms, total debt as a share of GDP rises from 33½ percent in 2018 to a projected 43½ percent in 2029 and 46 percent in 2039, remaining below the threshold of 55 percent for medium-capacity countries including Madagascar. The PV of debt to revenue and grants rises more moderately since the share of tax revenue in GDP is projected to continue rising off its low base. Despite this, and consistent with a shift to less concessional external financing sources and from external to more expensive domestic financing, the rise in debt service to revenue and grants increases toward the end of the projection horizon.

20. Total public debt is vulnerable to growth shocks and commodity prices (Figure 2; Table 4). The most severe test is the simulated GDP shock, where growth in 2020 and 2021 is one standard deviation lower than its historical average or the forecast (whichever yields lower GDP). For this shock, the PV of debt to GDP exceeds 55 percent of GDP by 2024, reaches 66 percent of GDP in 2029, and keeps rising. The threshold for medium-capacity countries like Madagascar is 55. Moreover, such a shock would also result in a debt service to revenue ratio above 90 percent before 2029 (and above 100 in the 2030s). Although there is no mechanical threshold, these ratios point to potentially severe debt service difficulties in the long run if no action to contain borrowing is taken soon after the shock. The assumed commodity shock would also breach the threshold for the PV of debt to GDP, reaching 64 percent in 2029, and raise the debt service to revenue ratio to 82 in the same year. An export shock, a natural disaster shock, or the contingent liability shock would also increase total public debt burdens.

[11] For example, for a threshold of 30 for the PV of debt to GDP, countries' whose baseline debt exceeds 24 under the baseline are assessed as having limited space. Madagascar's baseline PV exceeds that value over the medium term.

RISK RATING AND VULNERABILITIES

21. Madagascar is classified as being at low risk of external debt distress. Under the baseline, no thresholds are breached. Stress tests produce no breaches of the higher thresholds that apply following Madagascar's upgraded capacity to carry debt, although a shock to exports typically has the biggest effect on external PPG debt sustainability indicators and comes close. There are multiple breaches of the lower thresholds for countries with weak capacity.

22. The overall assessment is that Madagascar is at moderate risk of debt distress. Overall PPG debt stress tests, including to GDP growth and export commodity prices, breach the applicable threshold for the PV of debt to GDP as soon as 2024 and the ratio continues to rise. Together with high debt service burdens and Madagascar's relatively recent debt-carrying capacity upgrade, this prompts staff to assess Madagascar's overall risk of debt distress as moderate.

23. These assessments continue to be supportive of Madagascar's current plans to scale up its borrowing to meet its investment needs, though other factors are also critical. A steep increase in borrowing in line with a rapid execution of the government's ambitious medium-term borrowing plan would carry significant risks. There are countries in the region that have experienced sharp deterioration in their external debt distress rating from low to high within a short period of time. In addition to debt sustainability, other crucial considerations for the pace of borrowing include the economy's general absorptive capacity, public financial management, and public investment management.

24. Relevant factors that could affect future assessments include data revisions and the speed of realization of borrowing plans. Staff have assumed a slower pace of project disbursement than the authorities. Further shortfalls in disbursements on borrowing or alternatively more ambitious borrowing plans beyond the medium term would affect the debt profile. The state of SOE liabilities could also influence future assessments. Less grant financing and a switch to a less concessional mix of borrowing would raise the debt burden, especially when measured in PV terms, as well as debt service risks. Rebased (higher) GDP data is expected to be incorporated in the macroeconomic framework in the near future,[12] while efforts to enhance external statistics could improve private debt coverage. Madagascar's ability to preserve and build on its debt-carrying capacity is also important.

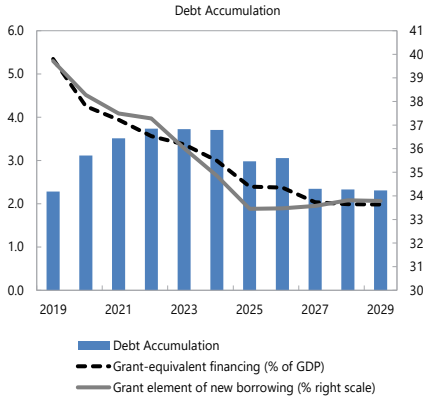
[12] New data according to the 1993 System of National Accounts (SNA93) with 2007 as the base year will be used by the authorities starting with the preparation of the 2020 budget. The rebased GDP in nominal terms is about 16 percent higher for the 2007-2017 period.

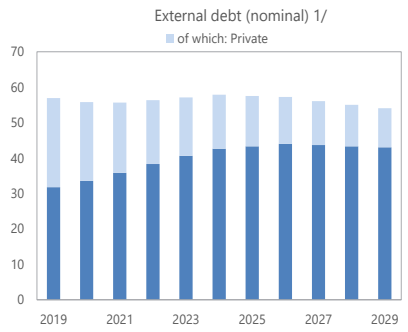
Table 1. Madagascar: External Debt Sustainability Framework, Baseline Scenario, 2016-2039

(In percent of GDP; unless otherwise indicated)

	Actual			Projections								Average 8/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
External debt (nominal) 1/	**59.7**	**57.7**	**58.0**	**56.9**	**55.8**	**55.6**	**56.3**	**57.1**	**57.9**	**54.1**	**46.9**	**45.6**	**56.3**
of which: public and publicly guaranteed (PPG)	**29.6**	**29.5**	**30.5**	**31.8**	**33.6**	**35.8**	**38.3**	**40.6**	**42.6**	**43.0**	**37.8**	**28.7**	**40.0**
Change in external debt	-2.1	-2.0	0.3	-1.1	-1.1	-0.2	0.7	0.8	0.8	-0.9	-0.8		
Identified net debt-creating flows	**-7.6**	**-9.6**	**-7.9**	**-4.3**	**-2.5**	**-1.5**	**-1.1**	**-0.8**	**-0.8**	**-1.3**	**-1.3**	**-0.7**	**-1.6**
Non-interest current account deficit	**-1.0**	**0.0**	**-1.4**	**1.3**	**2.8**	**3.5**	**3.7**	**3.9**	**3.8**	**2.9**	**2.6**	**5.2**	**3.1**
Deficit in balance of goods and services	2.2	3.8	3.9	6.2	6.3	7.2	7.6	8.0	8.2	7.4	7.0	**8.9**	**7.4**
Exports	33.1	35.5	36.0	34.7	34.8	34.0	33.4	32.9	32.5	30.2	26.9		
Imports	35.3	39.4	40.0	40.9	41.1	41.2	41.0	40.9	40.7	37.6	34.0		
Net current transfers (negative = inflow)	-6.8	-6.4	-8.0	-7.1	-5.5	-5.4	-5.4	-5.5	-5.6	-5.3	-4.9	**-6.2**	**-5.6**
of which: official	-3.4	-2.8	-3.0	-3.9	-2.4	-1.9	-1.4	-1.2	-0.9	-0.7	-0.6		
Other current account flows (negative = net inflow)	3.6	2.6	2.7	2.2	1.9	1.7	1.5	1.3	1.2	0.8	0.4	**2.5**	**1.3**
Net FDI (negative = inflow)	**-4.5**	**-3.1**	**-4.1**	**-3.2**	**-3.2**	**-3.1**	**-3.1**	**-3.1**	**-3.1**	**-3.1**	**-3.1**	**-5.2**	**-3.1**
Endogenous debt dynamics 2/	**-2.2**	**-6.5**	**-2.4**	**-2.3**	**-2.1**	**-1.9**	**-1.7**	**-1.6**	**-1.5**	**-1.1**	**-0.8**		
Contribution from nominal interest rate	0.4	0.4	0.6	0.6	0.7	0.7	0.8	0.9	1.0	1.2	1.1		
Contribution from real GDP growth	-2.5	-2.3	-2.8	-2.9	-2.8	-2.7	-2.5	-2.5	-2.5	-2.3	-1.9		
Contribution from price and exchange rate changes	-0.1	-4.7	-0.1	...	...	...	...	...	...	...	...		
Residual 3/	**5.5**	**7.6**	**8.3**	**3.2**	**1.4**	**1.4**	**1.8**	**1.6**	**1.6**	**0.4**	**0.5**	**3.7**	**1.2**
of which: exceptional financing	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators													
PV of PPG external debt-to-GDP ratio	**...**	**...**	**18.1**	**19.6**	**20.9**	**22.6**	**24.4**	**26.1**	**27.7**	**29.6**	**27.2**		
PV of PPG external debt-to-exports ratio	**...**	**...**	**50.2**	**56.5**	**60.1**	**66.4**	**73.2**	**79.5**	**85.1**	**98.1**	**101.0**		
PPG debt service-to-exports ratio	**3.1**	**5.6**	**2.6**	**2.6**	**3.0**	**3.4**	**3.9**	**4.7**	**5.0**	**5.9**	**7.8**		
PPG debt service-to-revenue ratio	**9.3**	**16.7**	**7.9**	**7.1**	**8.3**	**8.8**	**9.9**	**11.6**	**11.9**	**12.3**	**14.0**		
Gross external financing need (Million of U.S. dollars)	-425.0	12.5	-487.0	292.9	523.3	668.7	764.7	850.4	863.6	862.3	1470.3		
Key macroeconomic assumptions													
Real GDP growth (in percent)	4.2	4.3	5.2	5.2	5.3	5.1	4.9	4.8	4.8	4.6	4.3	**2.3**	**4.9**
GDP deflator in US dollar terms (change in percent)	0.2	8.6	0.3	-1.3	3.3	2.8	2.7	2.8	2.8	2.8	2.8	**0.5**	**2.5**
Effective interest rate (percent) 4/	0.7	0.8	1.1	1.1	1.3	1.5	1.6	1.8	2.0	2.4	2.6	**0.9**	**1.9**
Growth of exports of G&S (US dollar terms, in percent)	7.2	21.6	7.0	0.0	9.0	5.7	5.7	6.2	6.6	6.0	6.0	**6.3**	**5.7**
Growth of imports of G&S (US dollar terms, in percent)	3.3	26.2	7.1	6.3	9.3	8.3	7.2	7.5	7.3	6.4	6.2	**0.9**	**6.9**
Grant element of new public sector borrowing (in percent)	...	...	...	39.7	38.3	37.5	37.3	36.0	34.8	33.8	33.4	**...**	**35.6**
Government revenues (excluding grants, in percent of GDP)	11.1	11.9	12.0	12.5	12.6	12.9	13.1	13.4	13.7	14.5	15.1	**10.6**	**13.6**
Aid flows (in Million of US dollars) 5/	606.2	606.6	634.9	758.8	828.6	903.2	954.9	1013.1	999.1	1062.5	1796.3		
Grant-equivalent financing (in percent of GDP) 6/	...	...	...	5.3	4.3	3.9	3.6	3.4	3.0	2.0	1.7	**...**	**3.1**
Grant-equivalent financing (in percent of external financing) 6/	...	...	...	70.7	58.5	53.5	49.0	46.6	43.7	43.6	43.6	**...**	**49.0**
Nominal GDP (Million of US dollars)	10,126	11,464	12,090	12,559	13,662	14,765	15,901	17,132	18,457	26,646	54,259		
Nominal dollar GDP growth	4.4	13.2	5.5	3.9	8.8	8.1	7.7	7.7	7.7	7.5	7.2	**2.8**	**7.5**
Memorandum items:													
PV of external debt 7/	...	...	45.6	44.8	43.1	42.4	42.5	42.6	43.0	40.6	36.3		
In percent of exports	...	...	126.4	129.0	124.1	124.8	127.2	129.6	132.0	134.7	134.8		
Total external debt service-to-exports ratio	3.8	9.0	4.1	12.3	12.1	12.3	12.6	12.8	12.2	11.3	12.0		
PV of PPG external debt (in Million of US dollars)			2186.9	2462.9	2854.3	3334.1	3885.9	4477.9	5113.1	7880.9	14769.8		
(PVt-PVt-1)/GDPt-1 (in percent)				2.3	3.1	3.5	3.7	3.7	3.7	2.3	1.5		
Non-interest current account deficit that stabilizes debt ratio	1.1	2.1	-1.7	2.4	3.9	3.7	3.0	3.1	3.0	3.8	3.4		

Definition of external/domestic debt	Currency-based
Is there a material difference between the two criteria?	No





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

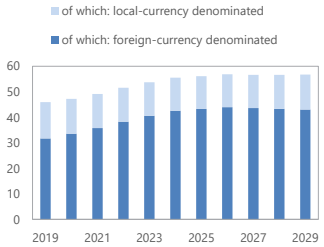
Table 2. Madagascar: Public Sector Debt Sustainability Framework, Baseline Scenario, 2016-2039

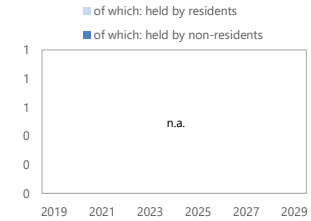
(In percent of GDP; unless otherwise indicated)

	Actual			Projections								Average 6/	
	2016	2017	2018	2019	2020	2021	2022	2023	2024	2029	2039	Historical	Projections
Public sector debt 1/	**47.1**	**46.0**	**45.7**	**46.0**	**47.2**	**49.2**	**51.6**	**53.7**	**55.6**	**56.8**	**56.1**	**42.3**	**53.3**
of which: external debt	**29.6**	**29.5**	**30.5**	**31.8**	**33.6**	**35.8**	**38.3**	**40.6**	**42.6**	**43.0**	**37.8**	**28.7**	**40.0**
Change in public sector debt	-3.5	-1.1	-0.3	0.3	1.3	2.0	2.4	2.1	1.8	0.1	-0.4		
Identified debt-creating flows	**-3.9**	**-3.8**	**-2.4**	**-0.9**	**1.1**	**1.9**	**2.5**	**2.4**	**2.1**	**0.1**	**-0.4**	**-0.2**	**1.0**
Primary deficit	**0.4**	**1.6**	**0.6**	**1.3**	**3.5**	**4.3**	**4.7**	**4.7**	**4.5**	**2.4**	**1.5**	**1.6**	**3.4**
Revenue and grants	14.6	14.8	14.8	16.4	15.0	14.8	14.4	14.6	14.6	15.2	15.7	**12.7**	**15.0**
of which: grants	3.4	2.9	2.8	3.9	2.4	1.9	1.4	1.2	0.9	0.7	0.6		
Primary (noninterest) expenditure	15.0	16.4	15.5	17.7	18.5	19.1	19.2	19.3	19.1	17.6	17.2	**14.2**	**18.4**
Automatic debt dynamics	**-4.3**	**-5.4**	**-3.0**	**-2.9**	**-3.0**	**-2.9**	**-2.8**	**-2.8**	**-2.9**	**-2.8**	**-2.4**		
Contribution from interest rate/growth differential	-3.6	-3.2	-3.7	-2.9	-2.8	-2.6	-2.4	-2.5	-2.6	-2.5	-2.2		
of which: contribution from average real interest rate	-1.5	-1.3	-1.4	-0.7	-0.4	-0.3	-0.2	-0.2	-0.2	-0.1	0.2		
of which: contribution from real GDP growth	-2.0	-1.9	-2.3	-2.3	-2.3	-2.3	-2.3	-2.4	-2.5	-2.5	-2.3		
Contribution from real exchange rate depreciation	-0.7	-2.2	0.7	...	...	...	...	...	...	...	...		
Other identified debt-creating flows	**0.0**	**0.0**	**0.0**	**0.6**	**0.6**	**0.6**	**0.5**	**0.5**	**0.5**	**0.5**	**0.5**	**0.0**	**0.5**
Privatization receipts (negative)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.6	0.6	0.6	0.5	0.5	0.5	0.5	0.5		
Residual	**0.4**	**2.7**	**2.1**	**1.2**	**-0.1**	**-0.3**	**-0.4**	**-0.5**	**-0.6**	**-0.3**	**-0.3**	**2.0**	**-0.2**
Sustainability indicators													
PV of public debt-to-GDP ratio 2/	**...**	**...**	**34.0**	**34.1**	**34.9**	**36.3**	**38.0**	**39.5**	**40.9**	**43.6**	**45.8**		
PV of public debt-to-revenue and grants ratio	**...**	**...**	**229.5**	**208.2**	**232.7**	**244.7**	**263.2**	**270.7**	**280.2**	**287.2**	**292.0**		
Debt service-to-revenue and grants ratio 3/	**7.0**	**7.0**	**43.5**	**39.4**	**45.6**	**51.6**	**56.2**	**59.9**	**62.0**	**70.0**	**96.7**		
Gross financing need 4/	1.5	2.7	7.1	8.9	11.1	12.6	13.4	14.0	14.1	13.6	17.2		
Key macroeconomic and fiscal assumptions													
Real GDP growth (in percent)	4.2	4.3	5.2	5.2	5.3	5.1	4.9	4.8	4.8	4.6	4.3	**2.3**	**4.9**
Average nominal interest rate on external debt (in percent)	0.8	0.9	1.1	1.1	1.4	1.4	1.4	1.5	1.5	1.6	1.6	**0.7**	**1.5**
Average real interest rate on domestic debt (in percent)	-7.8	-6.1	-6.8	-3.7	-1.6	-0.8	0.1	0.2	0.1	1.0	1.9	**-6.7**	**-0.2**
Real exchange rate depreciation (in percent, + indicates depreciation)	-2.5	-7.7	2.4	...	...	...	...	...	...	...	...	**1.1**	**...**
Inflation rate (GDP deflator, in percent)	8.5	6.5	7.3	6.7	6.3	5.8	5.2	5.0	5.0	5.0	5.0	**7.2**	**5.3**
Growth of real primary spending (deflated by GDP deflator, in percent)	8.5	14.0	-0.7	20.5	10.3	8.4	5.2	5.4	4.0	4.7	3.3	**1.9**	**6.3**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	3.9	2.7	0.9	1.1	2.3	2.3	2.3	2.5	2.7	2.4	1.9	**2.5**	**2.4**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Currency-based**
Is there a material difference between the two criteria?	**No**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central government, central bank, government-guaranteed debt, non-guaranteed SOE debt . Definition of external debt is Currency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

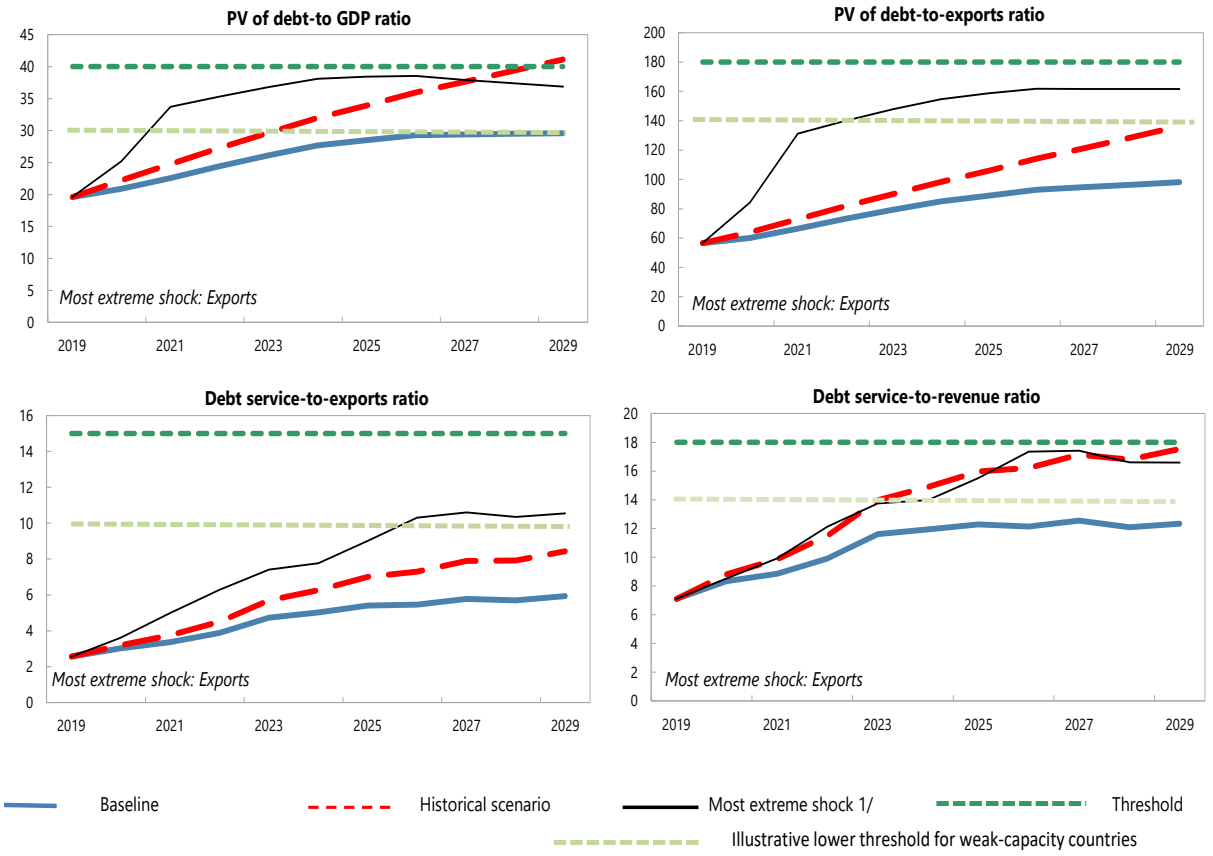
3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Figure 1. Madagascar: Indicators of Public and Publicly Guaranteed External Debt Under Alternatives Scenarios, 2019-2029



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	No	
Natural disaster	No	No
Commodity price	Yes	Yes
Market financing	n.a.	n.a.

Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

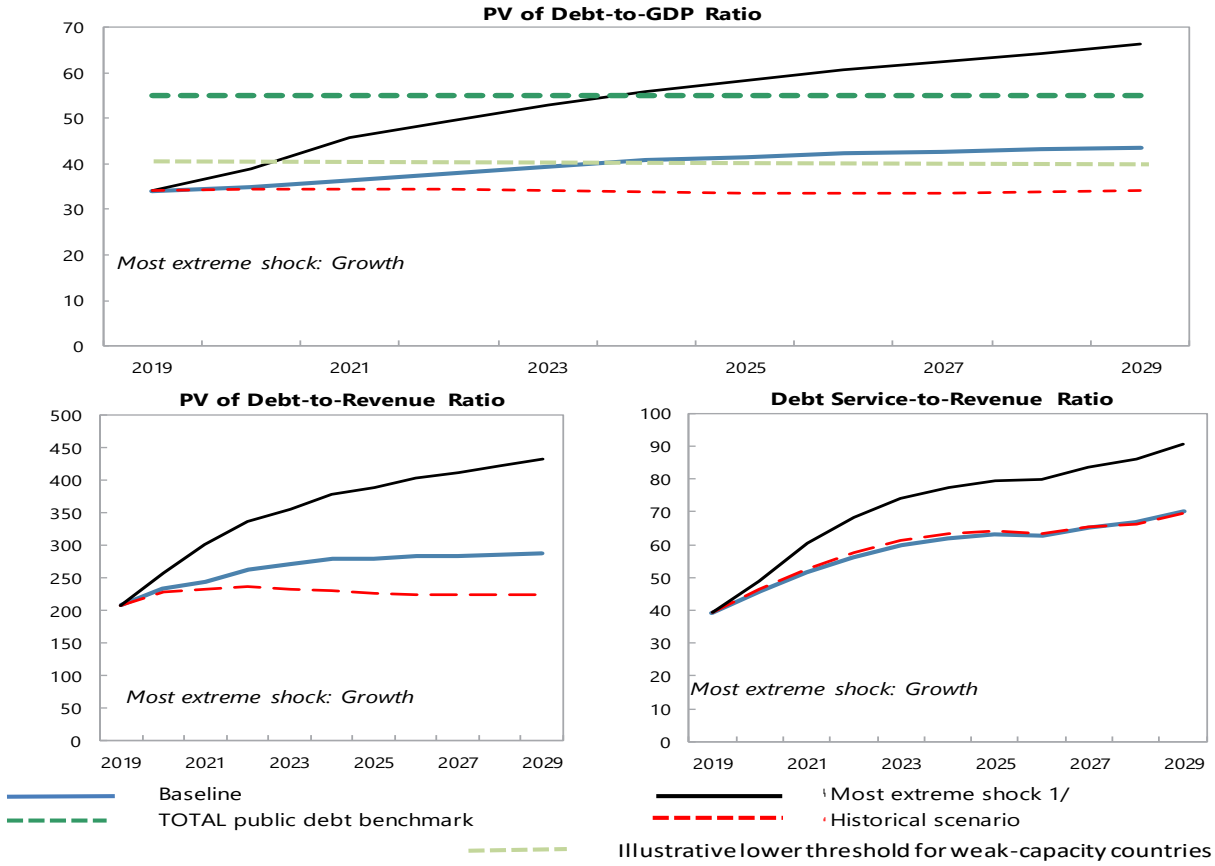
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	4	4

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Madagascar: Indicators of Public Debt Under Alternative Scenarios, 2019-2029



Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	38%	80%
Domestic medium and long-term	13%	10%
Domestic short-term	48%	10%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	1.6%	1.6%
Avg. maturity (incl. grace period)	26	26
Avg. grace period	4	4
Domestic MLT debt		
Avg. real interest rate on new borrowing	2.7%	3.5%
Avg. maturity (incl. grace period)	2	2
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	2.1%	3.0%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in or before 2029. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Table 3. Madagascar: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2019-2029

(In percent)

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of debt-to GDP ratio											
Baseline	20	21	23	24	26	28	28	29	29	29	30
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	20	22	25	27	30	32	34	36	38	39	**41**
B. Bound Tests											
B1. Real GDP growth	20	23	27	29	31	33	34	35	35	35	35
B2. Primary balance	20	21	23	25	27	28	29	30	30	30	30
B3. Exports	20	25	34	35	37	38	38	39	38	37	37
B4. Other flows 3/	20	23	26	28	30	31	32	32	32	32	32
B5. Depreciation	20	26	26	28	30	32	33	35	35	35	35
B6. Combination of B1-B5	20	27	31	33	35	37	37	38	38	38	38
C. Tailored Tests											
C1. Combined contingent liabilities	20	24	26	28	30	31	32	33	33	33	33
C2. Natural disaster	20	26	28	31	32	34	35	35	35	36	36
C3. Commodity price	20	22	24	26	28	29	30	30	30	30	30
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	40	40	40	40	40	40	40	40	40	40	40
PV of debt-to-exports ratio											
Baseline	57	60	66	73	79	85	89	93	95	96	98
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	57	64	73	82	90	98	106	114	121	129	136
B. Bound Tests											
B1. Real GDP growth	57	60	66	73	79	85	89	93	95	96	98
B2. Primary balance	57	61	68	75	82	87	91	95	97	98	100
B3. Exports	57	84	131	140	148	155	159	162	162	162	162
B4. Other flows 3/	57	66	78	85	91	96	99	103	104	105	106
B5. Depreciation	57	60	60	67	73	79	83	88	90	92	94
B6. Combination of B1-B5	57	78	77	99	106	113	116	120	121	123	124
C. Tailored Tests											
C1. Combined contingent liabilities	57	70	77	84	91	96	100	104	105	107	108
C2. Natural disaster	57	77	85	94	100	106	110	115	117	118	120
C3. Commodity price	57	63	72	79	85	90	94	97	99	100	101
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	180	180	180	180	180	180	180	180	180	180	180
Debt service-to-exports ratio											
Baseline	3	3	3	4	5	5	5	5	6	6	6
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	3	3	4	5	6	6	7	7	8	8	8
B. Bound Tests											
B1. Real GDP growth	3	3	3	4	5	5	5	5	6	6	6
B2. Primary balance	3	3	3	4	5	5	5	6	6	6	6
B3. Exports	3	4	5	6	7	8	9	10	11	10	11
B4. Other flows 3/	3	3	3	4	5	5	6	6	7	6	7
B5. Depreciation	3	3	3	4	5	5	5	5	5	5	6
B6. Combination of B1-B5	3	3	4	5	6	6	7	7	8	8	8
C. Tailored Tests											
C1. Combined contingent liabilities	3	3	4	4	5	5	6	6	6	6	6
C2. Natural disaster	3	3	4	4	5	6	6	6	6	6	7
C3. Commodity price	3	3	3	4	5	5	6	6	6	6	6
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	15	15	15	15	15	15	15	15	15	15	15
Debt service-to-revenue ratio											
Baseline	7	8	9	10	12	12	12	12	13	12	12
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	7	9	10	11	14	15	16	16	17	17	18
B. Bound Tests											
B1. Real GDP growth	7	9	11	12	14	14	15	14	15	14	15
B2. Primary balance	7	8	9	10	12	12	12	12	13	12	13
B3. Exports	7	9	10	12	14	14	16	17	17	17	17
B4. Other flows 3/	7	8	9	11	12	13	13	14	14	14	14
B5. Depreciation	7	10	11	12	14	15	15	14	15	14	14
B6. Combination of B1-B5	7	9	11	13	14	15	17	16	17	16	16
C. Tailored Tests											
C1. Combined contingent liabilities	7	8	9	11	12	13	13	13	13	13	13
C2. Natural disaster	7	8	10	11	13	13	13	13	13	13	13
C3. Commodity price	7	9	9	10	12	13	13	13	13	13	13
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Threshold	18	18	18	18	18	18	18	18	18	18	18

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

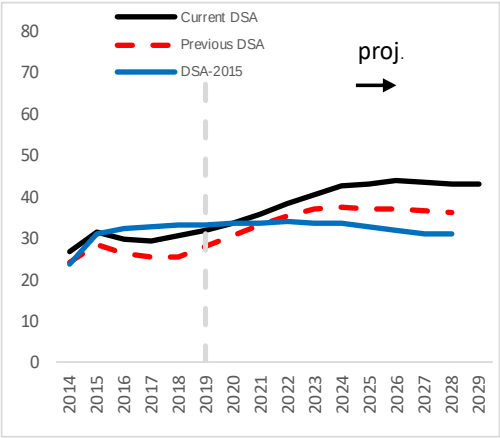
Table 4. Madagascar: Sensitivity Analysis for Key Indicators of Public Debt, 2019-2029

	Projections 1/										
	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029
PV of Debt-to-GDP Ratio											
Baseline	34	35	36	38	40	41	42	42	43	43	44
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	34	34	34	34	34	34	34	34	34	34	34
B. Bound Tests											
B1. Real GDP growth	34	39	46	50	53	**56**	**58**	**61**	**62**	**64**	**66**
B2. Primary balance	34	36	38	39	41	42	43	44	44	44	44
B3. Exports	34	39	46	48	49	50	50	50	50	50	49
B4. Other flows 3/	34	37	40	42	43	45	45	46	45	46	46
B5. Depreciation	34	38	38	38	38	38	38	38	37	37	37
B6. Combination of B1-B5	34	36	39	41	42	44	45	45	46	46	47
C. Tailored Tests											
C1. Combined contingent liabilities	34	40	41	42	44	45	46	46	46	47	47
C2. Natural disaster	34	43	44	45	47	48	49	49	50	50	50
C3. Commodity price	34	37	41	45	49	53	**55**	**58**	**60**	**62**	**64**
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
TOTAL public debt benchmark	55	55	55	55	55	55	55	55	55	55	55
PV of Debt-to-Revenue Ratio											
Baseline	208	233	245	263	271	280	280	285	285	286	287
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	208	229	232	237	233	231	226	224	223	224	225
B. Bound Tests											
B1. Real GDP growth	208	257	302	337	357	379	389	404	413	423	433
B2. Primary balance	208	238	255	273	280	289	288	292	291	292	293
B3. Exports	208	258	311	329	334	341	337	336	331	328	325
B4. Other flows 3/	208	246	271	290	296	305	303	305	303	302	302
B5. Depreciation	208	257	257	264	261	261	254	253	248	245	243
B6. Combination of B1-B5	208	239	259	281	289	300	300	304	304	305	306
C. Tailored Tests											
C1. Combined contingent liabilities	208	268	277	294	300	308	307	310	309	309	310
C2. Natural disaster	208	286	296	313	319	328	327	331	330	330	331
C3. Commodity price	208	248	279	317	340	364	375	388	398	408	419
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
Debt Service-to-Revenue Ratio											
Baseline	39	46	52	56	60	62	63	63	65	67	70
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2019-2029 2/	39	46	53	58	61	64	64	63	65	66	69
B. Bound Tests											
B1. Real GDP growth	39	49	60	68	74	77	79	80	84	86	91
B2. Primary balance	39	46	52	58	61	62	63	63	66	67	70
B3. Exports	39	46	52	58	61	63	66	67	69	70	73
B4. Other flows 3/	39	46	52	57	60	63	64	64	67	68	71
B5. Depreciation	39	44	50	55	59	61	62	61	63	65	67
B6. Combination of B1-B5	39	46	53	58	63	65	66	66	69	70	74
C. Tailored Tests											
C1. Combined contingent liabilities	39	46	57	61	61	63	64	63	66	67	71
C2. Natural disaster	39	46	60	65	63	65	66	65	67	69	72
C3. Commodity price	39	47	54	62	68	71	72	72	76	78	82
C4. Market Financing	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.

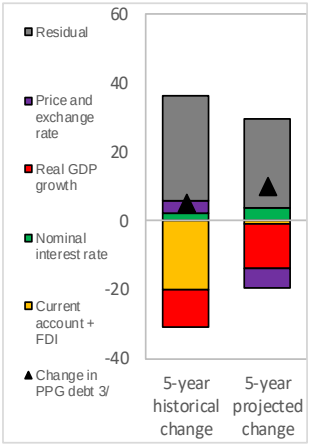
Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the benchmark.
2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.
3/ Includes official and private transfers and FDI.

Figure 3. Madagascar Debt Dynamics - Baseline Scenario: Drivers of External Debt

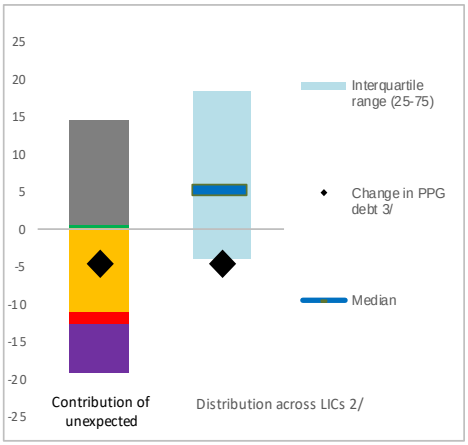
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

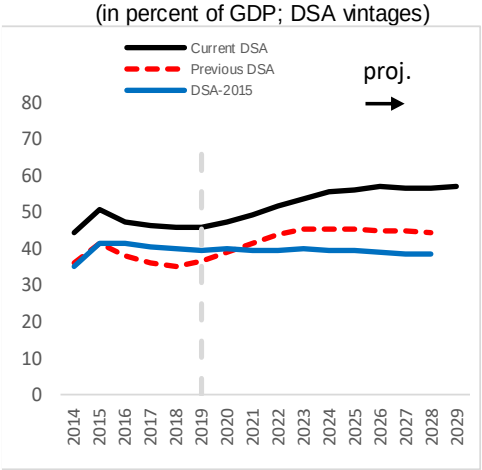


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

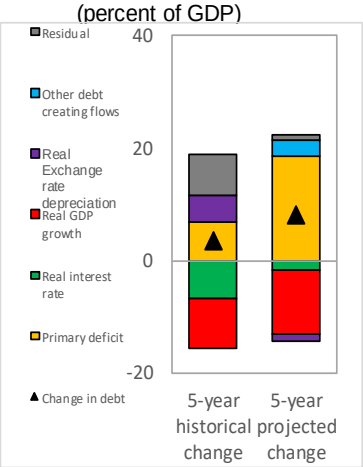


Public debt

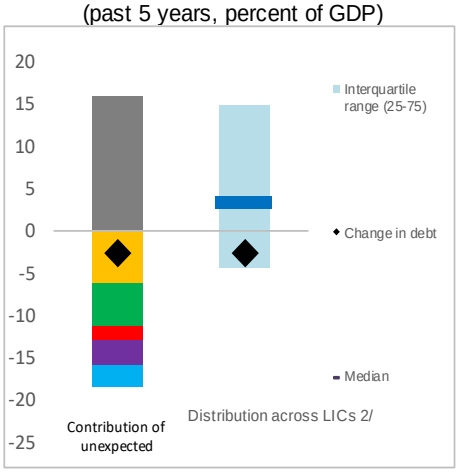
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



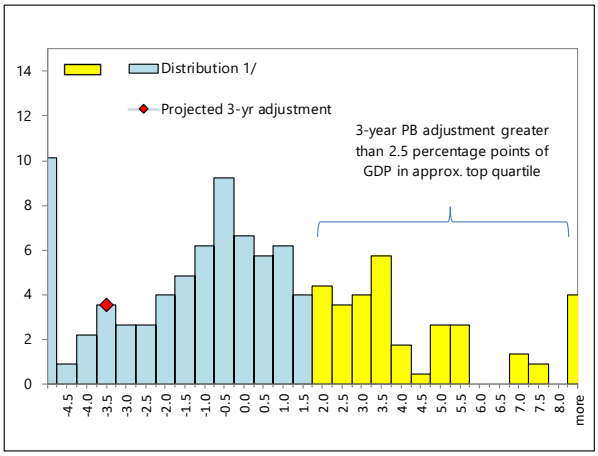
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

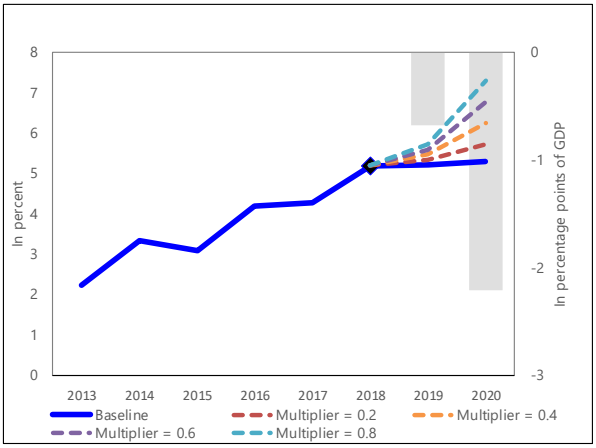
Figure 4. Madagascar: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



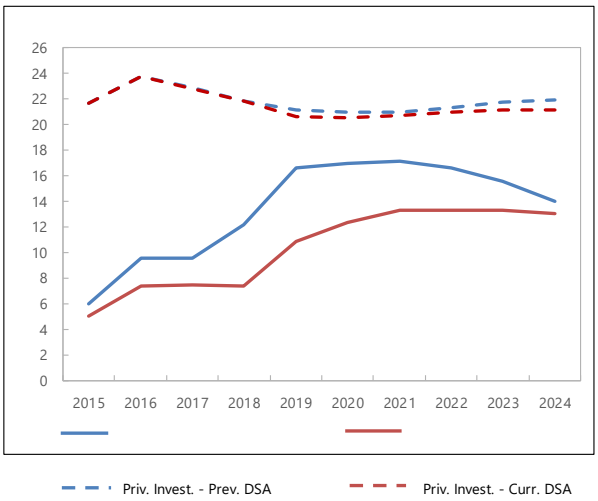
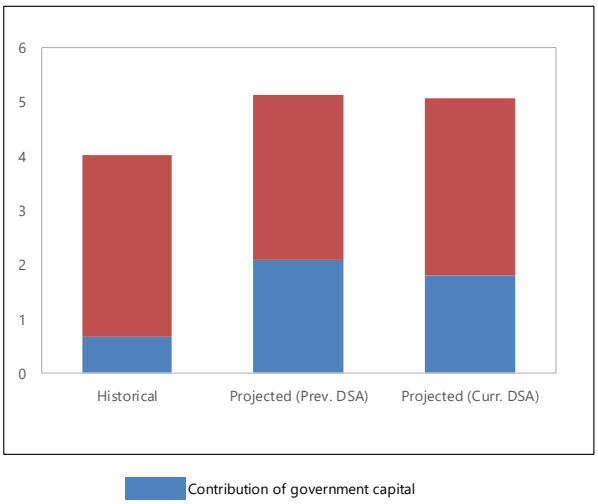
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/



1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)